

Retrieve, Summarize, and Verify: How Will ChatGPT Affect Information Seeking from the Medical Literature?

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JASN 34: 1302–1304, 2023. doi: <https://doi.org/10.1681/ASN.0000000000000166>

ChatGPT, a general purpose chatbot developed by OpenAI, has been widely reported to have the potential to revolutionize how people interact with information online.¹ Like other large language models (LLMs), ChatGPT has been trained on a large text corpus to predict probable words from the surrounding context. ChatGPT, however, has received substantial popular attention for generating human-like conversational responses, and new developments are occurring rapidly. The initial release in November 2022 used the model generative pretrained transformers (GPT)-3.5, and a version based on GPT-4.0 was released in March 2023. Recent work has discussed applications of ChatGPT for medical education and clinical decision support.^{2–4} However, health care professionals should be aware of the drawbacks and limitations—and potential capabilities—of using ChatGPT and similar LLMs to interact with medical knowledge.

A significant issue with ChatGPT is its tendency to generate confident-sounding but fabricated responses, commonly known as “hallucination.” Although it can generate plausible-sounding text, ChatGPT does not consult any source of truth.¹ Occasional incorrect or biased responses are, therefore, inevitable, and ChatGPT cannot faithfully cite sources to allow their evidence to be verified. For example, we prompted GPT-3.5-based ChatGPT to list possible mechanisms of AKI in

patients with coronavirus disease 2019 (COVID-19), with references. ChatGPT produced a list of plausible mechanisms, each of which has been investigated in the literature (Figure 1A) but supported its claims with fabricated titles and irrelevant PubMed identifiers. ChatGPT using the GPT-4 model acknowledges that it cannot produce references and refuses to cite articles in the answer (Figure 1C)—a significant improvement—but leaves the difficult task of verification to the user. Consulting ChatGPT directly for medical information carries a significant risk of drawing incorrect conclusions, particularly for patients.

However, LLMs such as ChatGPT may be better suited for text summarization than directly answering medical questions. Pilot studies have shown that LLMs can be effective at clinical reasoning, reading comprehension, and generating fluent summaries with high fidelity.^{5–7} Augmenting LLMs such as ChatGPT with a traditional literature search engine may therefore help address hallucination and enable the system to provide the user with comprehensive high-level summaries. For example, the new Bing provides ChatGPT-like interactive agents to help users create effective queries, identify relevant results, and summarize their content. In addition, ChatGPT recently introduced a plug-in that allows the model to search the web when generating responses (currently in limited beta test). Incorporating such

features into medical literature search has the potential to streamline gathering and synthesizing evidence from the literature, improving patient care.

Unfortunately, summaries generated by current LLMs such as ChatGPT also contain inaccuracies,⁸ in addition to practical issues such as limitations on the length of input text. To illustrate these problems, we prompted ChatGPT (model 3.5) to identify possible mechanisms of AKI in patients with COVID-19, using five manually chosen articles on the pathophysiology of AKI in COVID-19. Although ChatGPT’s summary (Figure 1B) generally reflects the mechanisms of AKI discussed by the articles, the response lists symptoms from only three of the five input studies and misses some important mechanisms mentioned repeatedly, such as hypoxia. Moreover, some citations are missing, contradicting evidence is not mentioned, similar mechanisms are not correctly grouped, and one of the mechanisms listed is less specific than the source. We also prompted ChatGPT (model 4),

Received: March 29, 2023 **Accepted:** May 24, 2023
Published Online Ahead of Print: May 31, 2023.

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A List the possible mechanisms of acute kidney injury in COVID-19 patients. Each mechanism should cite the source article title and PMID.

fabricated

There are several proposed mechanisms of acute kidney injury (AKI) in COVID-19 patients. Here are a few:

1. Direct viral infection: SARS-CoV-2 can directly infect kidney cells leading to acute kidney injury.
Source: "Kidney injury and repair mechanisms in SARS-CoV-2 associated acute kidney injury," PMID: 34429735.
2. Cytokine storm: COVID-19 patients may experience a cytokine storm, leading to systemic inflammation and kidney damage.
Source: "COVID-19 and acute kidney injury: mechanisms and evidences," PMID: 33535052.
3. Renin-angiotensin-aldosterone system (RAAS) dysregulation: SARS-CoV-2 can dysregulate the RAAS, leading to kidney injury.
Source: "COVID-19 and the Renin-Angiotensin System: Implications for Renal Involvement," PMID: 32618219.

(truncated response)

B

5 relevant [PMID, title, abstract]
32680910, 32660970, 32691342, 34226718, 33022818

List the possible mechanisms of acute kidney injury in COVID-19 patients. Each mechanism should cite the source PMID.

missing contradiction combination error not specific missing citation

1. Direct viral-induced tubular or glomerular injury (PMID: 32691342)
2. Sepsis-associated AKI (PMID: 32691342)
3. Thrombotic disease (PMID: 32691342)
4. Systemic haemodynamic instability contributing to tubular injury (PMID: 34226718)
5. Tissue inflammation and local immune cell infiltration (PMID: 34226718)
6. Endothelial injury and microvascular thrombi (PMID: 34226718)
7. Impaired type I interferon response (PMID: 34226718)
8. Dysregulation of immune responses including cytokine storm, macrophage activation syndrome, and lymphopenia (PMID: 33022818)
9. Organ interactions (PMID: 33022818)

(truncated response)

C List possible mechanisms of acute kidney injury in COVID-19 patients. Each mechanism should cite the source article title and PMID.

difficult to verify avoiding hallucination

I'm sorry for the misunderstanding, but as an AI model, I don't have the ability to access real-time databases or the internet to retrieve specific articles or their PMID numbers. However, I can provide a list of possible mechanisms of acute kidney injury (AKI) in COVID-19 patients based on my training data up to September 2021:

1. Direct viral injury: SARS-CoV-2 virus can directly infect kidney cells as they have abundant ACE2 receptors which the virus uses for cell entry. This can lead to direct cellular injury and death.
2. Cytokine storm: Severe COVID-19 infection can trigger a hyperactive immune response leading to a "cytokine storm". This can cause widespread inflammation and organ damage including the kidneys.
3. Hypoxia: Severe COVID-19 often causes significant respiratory compromise, leading to systemic hypoxia. Hypoxia can injure the kidneys and other organs.
4. Hemodynamic instability: Critically ill COVID-19 patients may experience periods of low

(truncated response)

D List possible mechanisms of acute kidney injury in COVID-19 patients. Each mechanism should cite the source article title and PMID.

uncomprehensive browsing the Web

Finished browsing

Searched: "mechanisms of acute kidney injury" Click failed

Searched: "PMID of Pathophysiology of COVID-19" Click failed

Searched: "PMID: 33060844" Clicked on: Reading content

From the articles, the following are possible mechanisms of acute kidney injury in COVID-19 patients:

1. Direct viral-induced tubular or glomerular injury is common in patients with COVID-19, potentially due to stability, which contributes to tubular injury. Notably, the injury is mild despite markedly reduced kidney function¹.

2. Tissue inflammation and local immune cell infiltration: These have been repeatedly reported to play a critical role in kidney injury. The inflammation is associated with damage-associated molecular patterns (DAMPs) and pathogen-associated molecular patterns (PAMPs) released locally within the lung and kidney, which recruit immune cells, including macrophages, effector T cells, and neutrophils, also contribute to the inflammation and tissue damage².

(truncated response)

Figure 1. Seeking information related to COVID-19 from ChatGPT. (A) ChatGPT-3.5 fabricates the references for its claims. (B) When prompted with the title and abstract of relevant articles, ChatGPT-3.5 can summarize the aspects requested, including some valid in-line references. However, many details are inaccurate: Related mechanisms are not correctly grouped, contradicting evidence is not mentioned, some citations are missing, and a listed mechanism is not specific. Note that the complete response mentioned only three of five articles. (C) ChatGPT-4 acknowledges its drawback and refuses to cite source articles in the response, which makes the verification more difficult. (D) ChatGPT-4 with the web browsing plug-in summarizes information from the search engine, but it only finds two relevant articles and fails to systematically answer the question. Responses were collected from GPT-3.5-based ChatGPT on March 14, 2023, and GPT-4-based ChatGPT on May 15, 2023.

with the web browsing plug-in, to answer the same question (Figure 1D). In this case, ChatGPT provides summaries with relevant citations. However, the response is based on only two articles, without synthesizing the evidence, and thus fails to answer the question systematically. Generally speaking, retrieval-augmented LLMs reduces users' control over sources, leading to results that may not align with the user's criteria for the best information (e.g., most up-to-date). Summaries generated by LLMs can miss important ideas discussed by the input sources or contain assertions unsupported by the input sources. Because imprecise or incomplete information can

lead to incorrect clinical decisions and patient harm, content generated by current LLMs, including ChatGPT, must be verified for accuracy and completeness before use. Further research on the faithfulness of LLM-generated summaries and automatic verification tools is also warranted to realize the full potential of LLMs to streamline evidence gathering and synthesis from the medical literature.

Despite these limitations, the remarkable progress in AI and the rapid advancements in LLMs may have profound implications for medicine, from assisting medical education,² to accelerating literature screening for systematic reviews,^{7,9}

and supporting physicians in various clinical settings as generalist medical AI,⁴ among other possibilities. In addition to augmenting LLMs with retrieval results from a search engine, LLMs can also be taught to use biomedical domain-specific tools, such as the Web APIs of specialized databases, to provide improved access to biomedical information.¹⁰

In summary, although ChatGPT is not currently suitable for use as a literature search engine on its own, we envision that the proposed retrieve, summarize, and verify paradigm could greatly benefit biomedical information seeking. This approach leverages the

impressive capability of LLMs to generate high-level summaries while minimizing the risk of directly using false or fabricated information by combining LLMs and search engines. It is important to note that although LLM technology is progressing rapidly, it has not yet matured enough for clinical use. Therefore, users must retain full responsibility for verifying the accuracy and reliability of its output.

DISCLOSURES

All authors have nothing to disclose.

FUNDING

This work was supported by the NIH Intramural Research Program, National Library of Medicine.

ACKNOWLEDGMENTS

This research was supported by the NIH Intramural Research Program, National Library of Medicine.

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